

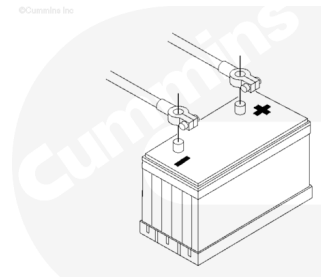
## Trainings ▾

### Preparatory Steps

#### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

- Disconnect the batteries.



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#### **⚠ WARNING ⚠**

**Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.**



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#### **⚠ WARNING ⚠**

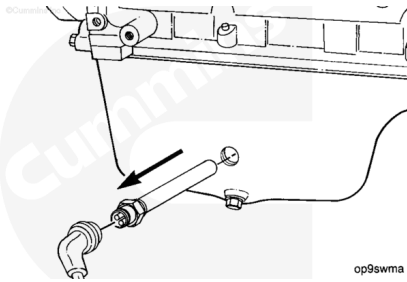
**To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.**

- Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>).

### Remove

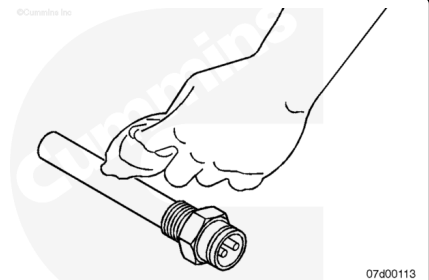
Disconnect the oil heater electrical cord.

Remove the heater element.

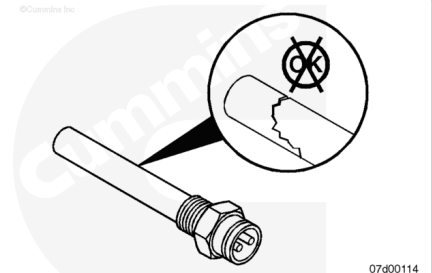


## Clean and Inspect for Reuse

Clean the heater element with a dry, clean towel.



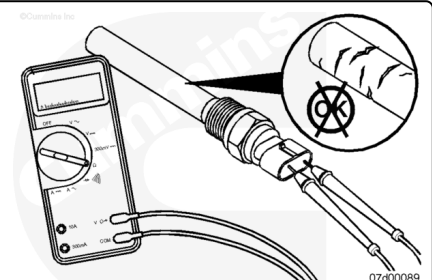
Check for cracks in the heating element.



## Test

Perform a resistance test on the engine lubricating oil heater element.

The readings **must** be a minimum of 91 ohms to a maximum 101 ohms.

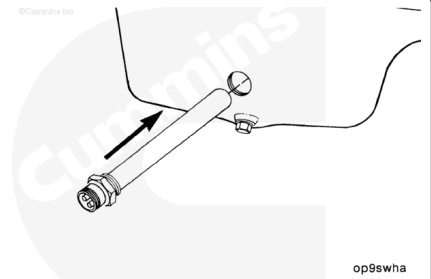


# Install

Replace the heater element.

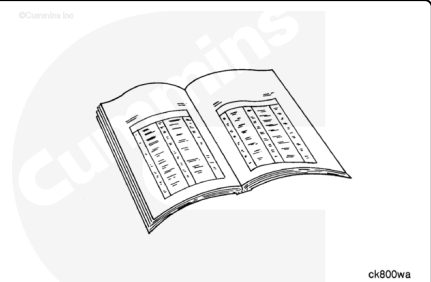
**Torque Value:** 80 N•m [ 59 ft-lb ]

Connect oil heater electrical cord.



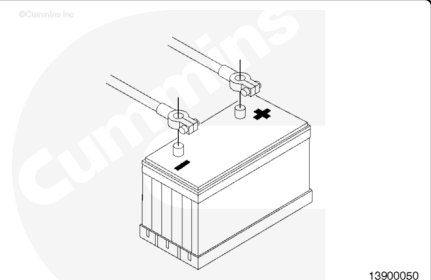
## Finishing Steps

- Refill the engine with clean oil to the correct level. Refer to Procedure 007-037 ([/qs3/pubsys2/xml/en/procedures/40/40-007-037.html](http://qs3/pubsys2/xml/en/procedures/40/40-007-037.html)).



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- Connect the batteries.

- Operate the engine and check for leaks.

